

ESVIL LABS

Esvil Protocol

*An Asset-Agnostic, Compliance-Aware Framework
for Real-World Asset Tokenization on BNB Chain*

David Nzagha

Esvil Labs

Version 2.0 · February 2026



*“Infrastructure for the ownership economy.
Tokenizing the world’s assets, one Diamond at a time.”*

Abstract

The tokenization of real-world assets (RWAs) promises to unlock trillions of dollars in illiquid value, yet current protocols remain asset-specific, architecturally rigid, and poorly suited to evolving regulatory requirements. The **Esvil Protocol** introduces a decentralized, compliance-aware infrastructure for tokenizing *any* real-world asset (including real estate, precious metals, commodities, intellectual property, and revenue-bearing instruments) on the BNB Chain. The protocol employs an **ERC-2535 Diamond Pattern** architecture, enabling modular, upgradeable smart contract systems where discrete functional facets (tokenization, compliance, trading, governance, revenue distribution) share a single proxy address and unified storage layer. A novel **E-RWA hybrid token standard** synthesizes ERC-3643 (T-REX compliant transfers), ERC-1643 (on-chain legal documents), ERC-1644 (controller-forced transfers), and ERC-1410 (partitioned balances into Class A, B, and C tranches) to accommodate diverse asset classes under a single composable framework. Community governance through vote-escrowed **veESVL** tokens controls protocol parameters, asset approvals, verifier elections, and facet upgrades. A decentralized network of staked Asset Verifiers provides due diligence attestations, with slashing penalties ensuring accountability. We present the protocol's formal architecture, tokenization flow, cryptoeconomic model, governance mechanics, and regulatory compliance strategy across multiple jurisdictions.

Keywords: real-world assets, RWA tokenization, ERC-2535 Diamond, BNB Chain, compliance, decentralized governance, fractional ownership, DeFi

Contents

1	Introduction	4
1.1	Background	4
1.2	The Esvil Thesis	4
1.3	Contributions	4
2	Related Work	5
2.1	Blockchain Foundations and Smart Contract Platforms	5
2.2	Asset Tokenization Frameworks	5
2.3	Security Token Standards and Compliance	8
2.4	Real-World Asset Identity and Authentication	8
2.5	Decentralized Finance and MSME Empowerment	9
2.6	Token Economics and Game Theory	10
2.7	Upgradeable Smart Contract Architecture	11
2.8	Summary of Gaps	11
3	Protocol Architecture	12
3.1	Design Rationale: Why the Diamond Pattern	12
3.2	Facet Architecture	13
3.2.1	Infrastructure Facets	13
3.2.2	Core Facets	13
3.3	Shared Storage: AppStorage	14
3.4	Upgrade Governance	15
4	The E-RWA Hybrid Token Standard	15
4.1	Design Philosophy	15
4.2	Standard Composition	15
4.3	Token Classes (Partitions)	16
4.4	Asset Type Configurations	16
4.5	Formal Token Model	16
5	Tokenization Flow	17
5.1	Phase 1: Asset Origination	17
5.2	Phase 2: Verification	17
5.3	Phase 3: Tokenization (Minting)	18
5.4	Phase 4: Distribution and Yield	18
5.4.1	Primary Distribution	18
5.4.2	Secondary Market	18
5.4.3	Revenue Distribution	18

6	Community Governance	19
6.1	Vote-Escrowed ESVL (veESVL)	19
6.2	Proposal Types	20
6.3	Decentralized Asset Verification Network	20
7	Token Economics	20
7.1	Token Utility	21
7.2	Token Allocation	21
7.3	Goal-Oriented Distribution	21
7.4	Deflationary Mechanics	22
8	Regulatory Compliance	23
8.1	Composable Compliance Modules	23
8.2	Supported Compliance Rules	23
8.3	zk-KYC and ONCHAINID	23
8.4	Jurisdictional Strategy	24
8.4.1	Liberland: Blockchain-Native Sovereignty	24
8.4.2	Burkina Faso: Tokenized Gold and Local Economic Empowerment	24
9	Use Cases	25
9.1	Real Estate: Fractional Commercial Ownership	25
9.2	Gold: Tokenized Vault Certificates	25
9.3	MSME Physical Spaces	25
9.4	Intellectual Property and Royalties	25
9.5	Agricultural Commodities	26
9.6	Burkina Faso: SONASP-Certified Artisanal Gold	26
10	Security and Risk Assessment	26
10.1	Smart Contract Security	26
10.2	Diamond-Specific Risks	27
10.3	Economic Risk Matrix	27
11	Implementation Roadmap	28
12	Conclusion	28

1. Introduction

1.1 Background

Real-world assets, spanning real estate, precious metals, commodities, fine art, intellectual property, and private credit, represent an estimated \$900 trillion in global value McKinsey [2024]. The vast majority of this value remains illiquid, fractionally inaccessible, and confined within opaque, jurisdiction-bound markets. Blockchain-based tokenization offers a paradigm shift: representing ownership rights as programmable tokens that can be fractionalized, traded, and composed within decentralized financial systems Zetsche et al. [2020], Schär [2021].

Despite significant advances, the current RWA tokenization landscape suffers from three structural deficiencies:

1. **Asset Specificity:** Most protocols are purpose-built for a single asset class (real estate, gold, carbon credits), requiring entirely new infrastructure for each new category.
2. **Architectural Rigidity:** Monolithic smart contract designs cannot adapt to evolving regulatory requirements, new asset types, or protocol improvements without costly redeployment.
3. **Compliance Fragmentation:** Regulatory compliance is either absent (limiting institutional participation) or hardcoded (limiting jurisdictional flexibility), with no standard framework for composable compliance modules.

1.2 The Esvil Thesis

The Esvil Protocol posits that RWA tokenization infrastructure should be *asset-agnostic*: a single composable protocol capable of onboarding, verifying, tokenizing, trading, and distributing yield from any real-world asset, governed entirely by its community.

This asset-agnostic approach is enabled by three architectural innovations:

1. The **ERC-2535 Diamond Pattern**, which organizes protocol logic into modular, independently upgradeable facets behind a single proxy address.
2. The **E-RWA hybrid token standard**, which synthesizes four ERC standards into a flexible container adaptable to any asset class.
3. A **pluggable compliance engine** that enforces jurisdiction-specific rules through composable compliance modules, enabling the same protocol to operate across Nigeria, Switzerland, Estonia, Singapore, Liechtenstein, Burkina Faso, and beyond.

1.3 Contributions

This paper makes the following contributions:

1. A formal specification of the Diamond-based protocol architecture for asset-agnostic RWA tokenization (Section 3).
2. The E-RWA hybrid token standard with multi-class partitioned ownership (Section 4).
3. A complete tokenization flow from asset origination through verification, minting, distribution, and yield (Section 5).

4. Community governance mechanics with veESVL voting and decentralized verifier networks (Section 6).
5. Cryptoeconomic model with goal-oriented distribution and deflationary mechanics (Section 7).
6. Multi-jurisdictional regulatory compliance strategy (Section 8).

2. Related Work

The Esvil Protocol builds upon a rich body of work spanning blockchain architecture, asset tokenization, decentralized governance, MSME financial inclusion, and compliance-aware token standards. This section surveys the state of the art across seven domains, identifying both foundational contributions and the gaps that motivate Esvil's design.

2.1 Blockchain Foundations and Smart Contract Platforms

The conceptual foundation for decentralized, trustless systems was laid by Nakamoto Nakamoto [2008], whose Bitcoin whitepaper introduced the proof-of-work consensus mechanism and the notion of a distributed, immutable ledger. Buterin Buterin [2014] extended this with Ethereum, introducing Turing-complete smart contracts that enabled programmable financial logic on-chain. Wood Wood [2014] formalized Ethereum's execution model in the Yellow Paper, specifying the Ethereum Virtual Machine (EVM), gas mechanics, and state transition functions that underpin all ERC token standards used in the Esvil Protocol.

Benton and Radziwill Benton and Radziwill [2017] provided an accessible primer for software quality professionals, disambiguating blockchain terminology and discussing practical utility beyond cryptocurrency, including supply chain provenance, digital identity, and certification. Their work underscored the importance of software quality assurance in blockchain deployments, a concern directly addressed by Esvil's multi-audit security model.

Szabo Szabo [1997] anticipated many of these developments with his formalization of "smart contracts" as cryptographic protocols that embed contractual clauses into hardware and software. His vision of contracts that "minimize the need for trusted intermediaries" is central to Esvil's design philosophy: compliance, governance, and revenue distribution encoded entirely in on-chain logic.

2.2 Asset Tokenization Frameworks

The tokenization of real-world assets has attracted substantial academic and industry attention. Konashevych Konashevych [2020] introduced the concept of "Title Tokens" for real estate, arguing for technological neutrality where blockchain-based registries coexist with traditional land registries. His framework illustrated how ownership tokens can be cryptographically linked to off-chain certificates (Figure 1), addressing the legal, technological, and organizational dimensions of real estate tokenization. He established that tokenization is not merely a technical exercise but requires coordinated reform across property law, registry governance, and citizen rights. Esvil's DocumentFacet (ERC-1643) and multi-jurisdictional compliance strategy directly extend this vision to a broader asset universe.

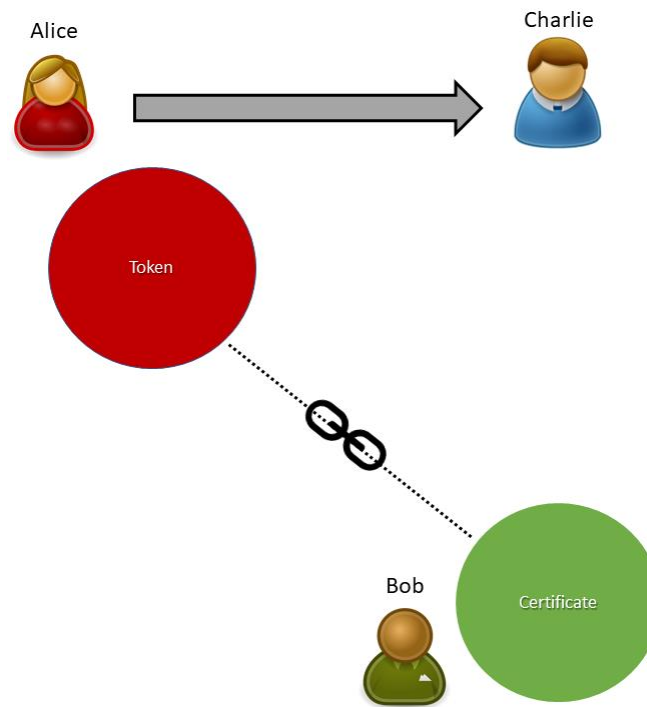


Figure 1. The Title Token model showing the cryptographic linkage between on-chain ownership tokens and off-chain certification authorities. Adapted from Konashevych Konashevych [2020].

Joshi and Choudhury Joshi and Choudhury [2022] proposed a blockchain-based framework for real estate tokenization using ERC-1155 tokens and IPFS for document storage. Their work identified the structural problems in traditional real estate markets (Figure 2) and demonstrated that tokenization can lower entry barriers, fix liquidity constraints, and improve stakeholder interaction. However, their framework was limited to a single asset class (real estate) and a single token standard, lacking the multi-class partition model and compliance composability that Esvil provides through the E-RWA hybrid standard.

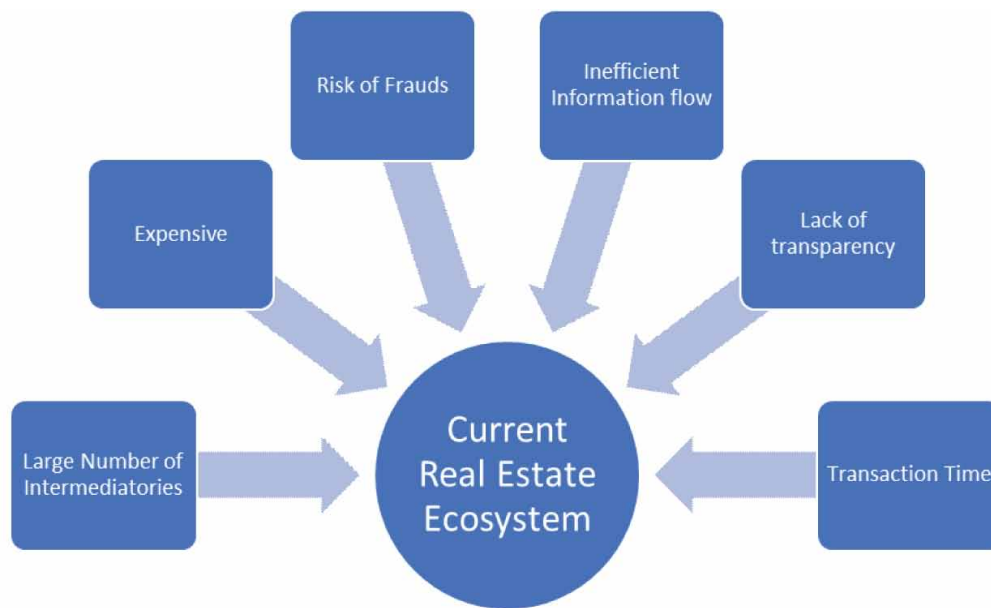


Figure 2. Structural problems in traditional real estate markets that tokenization aims to solve, including illiquidity, high entry barriers, and lack of transparency. Adapted from Joshi and Choudhury Joshi and Choudhury [2022].

Juan et al. [2023] surveyed tokenized markets broadly, examining architectures, designs, challenges, and best practices for tokenized platforms across digital asset categories. Their taxonomy of tokenizable assets (Figure 3) distinguishes between physical and digital assets, fungible and non-fungible representations, providing the conceptual foundation for Esvil's asset-agnostic approach. They identified four primary steps in tokenization (asset identification, valuation, parameterization, and smart contract creation) and highlighted cybersecurity threats and regulatory absence as key challenges. Esvil addresses both: the VerifierFacet provides decentralized valuation and due diligence, while the ComplianceFacet implements composable regulatory modules.

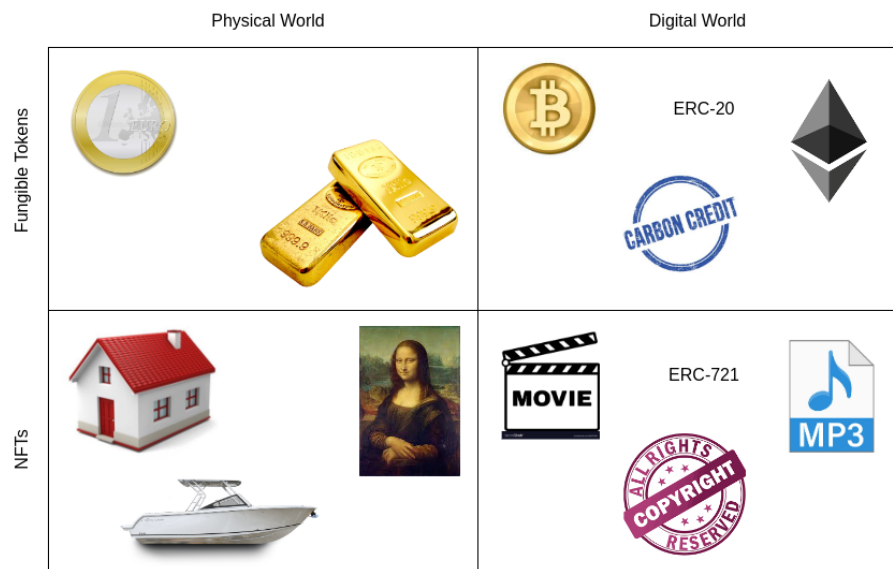


Figure 3. Taxonomy of tokenizable assets across physical and digital categories, illustrating the breadth of asset classes that a tokenization protocol must accommodate. Adapted from Juan et al. [2023].

The OECD [2022a] provided an institutional perspective on asset tokenization, analyzing how tokenization of financial instruments could improve market efficiency, reduce settlement times, and democratize access to investment products. Their report emphasized that regulatory clarity, investor protection, and interoperability standards are prerequisites for mainstream adoption, all of which inform Esvil’s jurisdictional strategy and compliance architecture.

2.3 Security Token Standards and Compliance

The ERC-3643 (T-REX) standard [2021] established the paradigm for compliance-aware token transfers, introducing modular compliance agents that enforce transfer restrictions based on investor identity and jurisdictional rules. Edmonds et al. [2018] proposed the ERC-1400 family (ERC-1410 for partitioned balances, ERC-1643 for document management, ERC-1644 for controller operations), providing the primitives for representing complex security instruments on-chain. Esvil’s E-RWA standard synthesizes all four of these standards into a single composable framework, an integration not previously achieved in any deployed protocol.

The Financial Cryptography and Data Security conference proceedings Baldimtsi and Cachin [2023] collected cutting-edge research on cryptographic protocols for financial systems, including work on privacy-preserving identity verification, DeFi liquidation mechanics, and formal security models for smart contracts. Esvil’s zk-KYC integration draws from this body of work, employing zero-knowledge proofs for identity attestation without exposing personal data.

2.4 Real-World Asset Identity and Authentication

Norta, Udokwu, and Craß [2024] addressed a critical gap in RWA tokenization: the authentication of real-world asset identities in blockchain-enabled inter-organizational systems.

Their work combined a logistics DApp for RWA tracking with a multi-factor challenge-set self-sovereign identity authentication system (MFSSIA), using NFT-based identity attestations to prevent counterfeit circulation (Figure 4). Their configurable challenge-response framework for verifying asset authenticity across organizational boundaries directly informs Esvil’s VerifierFacet design, where staked verifiers submit signed attestations after multi-dimensional due diligence (legal, physical, and valuation).

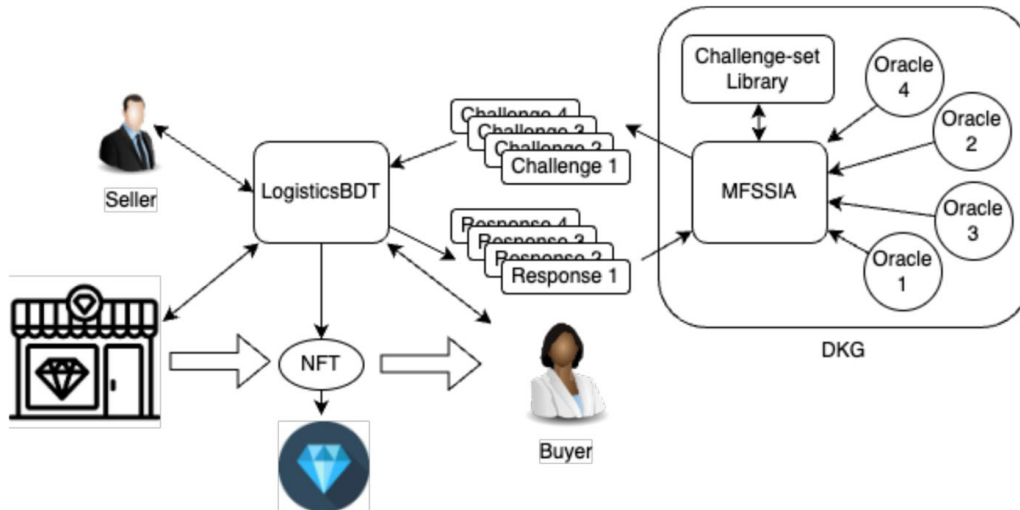


Figure 4. The Multi-Factor Challenge-Set Self-Sovereign Identity Authentication (MFSSIA) system for real-world asset identity verification in blockchain-enabled inter-organizational systems. Adapted from Norta et al. Norta et al. [2024].

Karamachoski, Marina, and Taskov Karamachoski et al. [2020] demonstrated blockchain-based certification management, applying Ethereum smart contracts to university diploma verification. While their application domain differs, the core architecture of on-chain certification with cryptographic hash verification and transparent auditability is directly applicable to Esvil’s DocumentFacet, which anchors legal documents (title deeds, assay certificates, audit reports) on-chain via content hashes stored on IPFS/Arweave.

2.5 Decentralized Finance and MSME Empowerment

The role of MSMEs in emerging economies has been extensively documented. The OECD [2022b] reported that SMEs account for over 90% of businesses and 60-70% of employment globally, yet face persistent barriers to finance, technology adoption, and market access. The World Bank [2023] highlighted that Africa’s MSME sector, despite its dynamism, is severely constrained by capital access, with an estimated \$330 billion financing gap across the continent.

Amey Navarro Amey Navarro [2024] investigated DeFi as a tool for MSME financial empowerment in Colombia, conducting expert interviews and MSME surveys to evaluate whether decentralized finance could address high interest rates, slow loan approvals, and limited access to capital. His findings confirmed DeFi’s potential but identified knowledge gaps, regulatory uncertainty, and infrastructure limitations as barriers, challenges that Esvil addresses through its asset-agnostic tokenization framework, which enables MSMEs to unlock value from physi-

cal assets (commercial spaces, equipment, inventory) without relying on traditional banking infrastructure.

Riggs, Vyas, and Sethi Riggs et al. [2022] explored blockchain and Distributed Autonomous Community (DAC) ecosystems for democratizing finance and delivery of transport, housing, and urban infrastructure. Their work demonstrated how tokenized community assets could enable fractional ownership, transparent governance, and automated revenue distribution, a model closely aligned with Esvil's MSME physical spaces use case, where cooperatives co-own commercial spaces through E-RWA Class B tokens.

Akaba, Norta, Udokwu, and Draheim Akaba et al. [2020] proposed a blockchain-based framework for e-procurement in Nigeria's public sector, addressing corruption, lack of transparency, and weak documentation systems. Their framework for on-chain interoperability, citizen participation, and project auditability provides relevant context for Esvil's deployment strategy in Nigeria, where regulatory compliance and institutional trust are critical adoption factors.

2.6 Token Economics and Game Theory

Udokwu Udokwu [2024] formalized the token economics of a blockchain-based research collaboration platform using game theory, employing extensive-form games to analyze token distribution fairness and designing a logarithmic utility pricing model for token sustainability. His work demonstrated that self-sustaining token models require internal utility backing (not purely speculative value) and dynamic pricing mechanisms. Esvil's ESVL token economics incorporate these insights: the goal-oriented distribution model ties token emissions to protocol KPIs (Total Value Tokenized, active verifiers, governance participation), and the deflationary burn mechanics ensure long-term supply reduction.

Schär Schär [2021] provided a comprehensive taxonomy of DeFi protocols, analyzing lending, decentralized exchanges, derivatives, and asset management on blockchain-based financial markets. His framework for understanding composability, the ability of DeFi protocols to interact and build upon each other, directly informs Esvil's Diamond Pattern architecture, where facets can be composed, upgraded, and extended without disrupting existing functionality.

Akbar et al. Akbar et al. [2025] designed and implemented a tokenized Decentralized Energy Management System (DEMS) for peer-to-peer energy trading, demonstrating how energy tokenization through custom tokens (ENRTokens) and smart contracts could automate trustless transactions. Their modular smart contract architecture (Figure 5) and microservices-based design for scalability across different hardware integrations parallel Esvil's facet-based modularity, though Esvil extends this pattern to arbitrary asset classes rather than a single commodity.

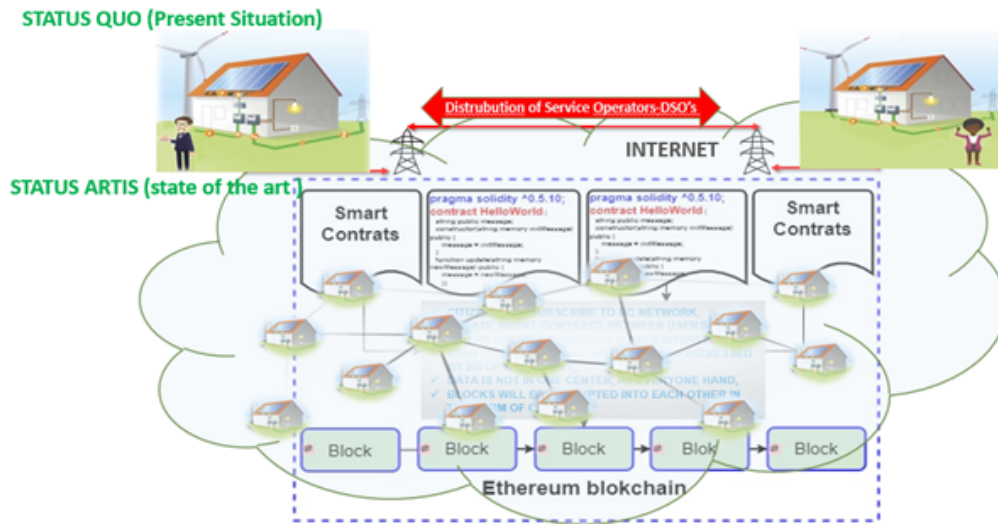


Figure 5. Architecture of the tokenized Decentralized Energy Management System (DEMS), illustrating modular smart contract design for peer-to-peer energy trading. Adapted from Akbar et al. [2025].

2.7 Upgradeable Smart Contract Architecture

The ERC-2535 Diamond standard Mudge [2020] introduced a paradigm shift in smart contract architecture by enabling unlimited functionality through modular facets behind a single proxy address. Unlike earlier proxy patterns (transparent proxies, UUPS), the Diamond pattern supports granular function-level upgrades, shared storage across facets, and full on-chain introspection. While the Diamond pattern has been adopted in protocols like Aavegotchi and Louper, its application to RWA tokenization, where regulatory requirements demand frequent compliance module updates and new asset types require new functionality, has not been systematically explored prior to this work.

The combination of ERC-2535 with compliance-aware token standards (ERC-3643) and partitioned balances (ERC-1410) within a single Diamond represents a novel architectural contribution. Previous RWA protocols have employed monolithic contracts or loosely coupled multi-contract systems, both of which impose upgrade friction and composability costs that the Diamond pattern eliminates.

2.8 Summary of Gaps

Table 1 summarizes the key limitations in existing work that the Esvil Protocol addresses.

Table 1. Summary of gaps in existing work addressed by the Esvil Protocol.

Gap	Existing Approaches	Esvil's Solution
Asset specificity	Separate protocols per asset class (real estate, energy, gold)	Asset-agnostic E-RWA standard configurable for any asset
Architectural rigidity	Monolithic contracts or simple proxy patterns	ERC-2535 Diamond with modular, independently upgradeable facets
Compliance fragmentation	Hardcoded rules or absent compliance	Pluggable compliance modules composable per jurisdiction
MSME exclusion	Traditional finance barriers, high entry costs	Fractional tokenization, cooperative ownership models
Verification centralization	Centralized appraisers and auditors	Decentralized staked verifier network with slashing
Token sustainability	Speculative token models without utility backing	Goal-oriented distribution tied to protocol KPIs, deflationary burns
Identity privacy	KYC requiring full data disclosure	zk-KYC via ONCHAINID preserving user privacy

3. Protocol Architecture

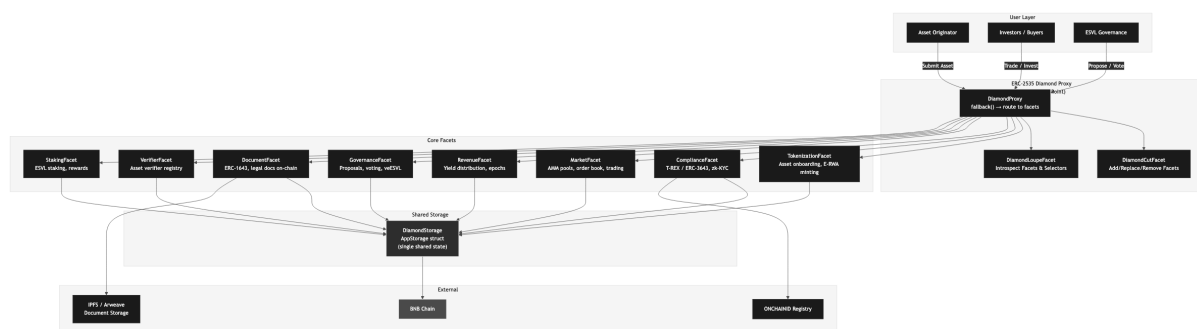


Figure 6. Esvil Protocol architecture. The ERC-2535 Diamond Proxy routes all calls through a single address to modular facets (Tokenization, Compliance, Market, Revenue, Governance, Document, Verifier, Staking), all sharing a unified AppStorage struct. External integrations include BNB Chain, IPFS/Arweave for document storage, and the ONCHAINID registry for identity verification.

3.1 Design Rationale: Why the Diamond Pattern

Traditional smart contract systems deploy monolithic contracts or loosely coupled proxy-delegate patterns. Both approaches impose critical limitations for RWA tokenization:

- **Size limits:** The EVM's 24KB contract size limit constrains the amount of logic deployable

in a single contract, forcing artificial splits.

- **Upgrade friction:** Transparent proxies and UUPS patterns require careful storage layout management and cannot add new state variables without migration risk.
- **Composability cost:** Multi-contract systems require cross-contract calls, increasing gas costs and attack surface.

The ERC-2535 Diamond standard Mudge [2020] resolves these constraints by introducing *facets*: independent contract modules whose functions are registered with a single Diamond proxy. The proxy's `fallback()` function routes calls to the appropriate facet based on function selector mappings.

Key properties:

- **No size limit:** The Diamond can incorporate unlimited functionality across unlimited facets.
- **Shared storage:** All facets read and write to a single `AppStorage` struct, eliminating cross-contract call overhead.
- **Granular upgrades:** Individual facets can be added, replaced, or removed without affecting others, through the `diamondCut()` function.
- **Single address:** Users and integrators interact with one contract address regardless of protocol complexity.
- **Introspection:** The `DiamondLoupeFacet` enables on-chain inspection of all registered facets and function selectors, ensuring full transparency.

3.2 Facet Architecture

The Esvil Diamond comprises eight core facets, two infrastructure facets, and a shared storage layer.

3.2.1 Infrastructure Facets

- **DiamondCutFacet:** Manages facet registration, replacement, and removal. Protected by governance timelock; only executable after successful veESVL vote and 48-hour delay.
- **DiamondLoupeFacet:** Implements ERC-2535 introspection interface. Returns all registered facets, their function selectors, and supported interfaces.

3.2.2 Core Facets

1. **TokenizationFacet:** Manages the asset onboarding pipeline. Receives verified asset metadata, mints E-RWA tokens with configurable partitions, and registers the asset in the global registry.
2. **ComplianceFacet:** Enforces transfer restrictions based on jurisdiction, investor accreditation, and holding limits. Integrates with the ONCHAINID registry for zk-KYC verification. Implements composable compliance modules that can be swapped per jurisdiction.
3. **MarketFacet:** Provides primary distribution (fixed-price offerings, Dutch auctions) and secondary market trading through integrated AMM pools. Manages liquidity provisioning and order matching.

4. **RevenueFacet:** Collects and distributes yield from tokenized assets (rental income, commodity sales, royalties) to E-RWA holders on a pro-rata basis during defined revenue-sharing epochs.
5. **GovernanceFacet:** Manages proposal submission, veESVL-weighted voting, quorum enforcement, timelock execution, and emergency actions. Controls all protocol parameter changes.
6. **DocumentFacet:** Implements ERC-1643 for attaching, updating, and retrieving legal documents (title deeds, assay certificates, audit reports) linked to tokenized assets. Documents are stored on IPFS/Arweave with content hashes anchored on-chain.
7. **VerifierFacet:** Manages the decentralized Asset Verifier network. Handles verifier applications, staking, assignment to assets, attestation submission, reputation tracking, and slashing.
8. **StakingFacet:** Manages ESVL token staking for verifiers, governance participants, and liquidity providers. Handles reward distribution, unbonding periods, and slashing execution.

3.3 Shared Storage: AppStorage

All facets share a single AppStorage struct via Solidity's diamond storage pattern. This eliminates storage collision risk and ensures atomic state transitions:

```
struct AppStorage {
    // Asset Registry
    mapping(bytes32 => Asset) assets;
    mapping(bytes32 => TokenConfig) tokenConfigs;
    uint256 totalAssetsRegistered;

    // Compliance
    mapping(address => Identity) identities;
    mapping(bytes32 => ComplianceRule[]) jurisdictionRules;

    // Market
    mapping(bytes32 => Pool) liquidityPools;
    mapping(bytes32 => Offering) offerings;

    // Revenue
    mapping(bytes32 => Epoch[]) revenueEpochs;
    mapping(bytes32 => mapping(address => uint256)) claimable;

    // Governance
    mapping(uint256 => Proposal) proposals;
    mapping(address => VeLock) veLocks;
    uint256 proposalCount;

    // Verifiers
    mapping(address => Verifier) verifiers;
    mapping(bytes32 => Attestation[]) attestations;
```

```
// Token
IERC20 esvlToken;
uint256 totalStaked;
}
```

Listing 1. AppStorage struct (simplified)

3.4 Upgrade Governance

Protocol upgrades follow a strict governance pipeline:

1. A veESVL holder submits a proposal specifying the `diamondCut()` parameters (facet address, action, function selectors).
2. The community votes over a 7-day period with a 10% quorum requirement and 67% approval threshold.
3. Approved proposals enter a 48-hour timelock before execution.
4. Emergency proposals (circuit breakers, critical patches) require 80% supermajority and 24-hour timelock.

This ensures that the protocol is both upgradeable and resistant to unilateral modification.

4. The E-RWA Hybrid Token Standard

4.1 Design Philosophy

The E-RWA token is a composable security token standard designed to represent ownership, access, and economic rights in any real-world asset. Rather than creating asset-specific token contracts, E-RWA provides a single, configurable token framework adaptable to diverse asset classes through parameterization.

4.2 Standard Composition

The E-RWA standard synthesizes four established ERC specifications:

Table 2. ERC standards composing the E-RWA hybrid token.

Standard	Function in E-RWA
ERC-3643 (T-REX)	Compliance-aware transfer restrictions, investor whitelisting, modular compliance agents
ERC-1643	On-chain attachment of legal documents (deeds, certificates, audit reports) to token contracts
ERC-1644	Controller-forced transfers for regulatory compliance, dispute resolution, and asset recovery
ERC-1410	Partitioned balances enabling multiple token classes with distinct rights per asset

4.3 Token Classes (Partitions)

Each tokenized asset issues E-RWA tokens partitioned into three classes via ERC-1410:

Table 3. E-RWA token class structure.

Class	Rights	Transferability
A (Ownership)	Voting + revenue share + access rights	Restricted (KYC-gated, compliance-checked)
B (Access)	Revenue share + access/usage rights, no voting	Semi-restricted (lighter KYC)
C (Speculative)	Tradeable derivative, exposure to asset value	Free trading on AMM pools

This partition model accommodates diverse investor profiles within a single asset:

- Institutional investors acquire Class A tokens for governance and yield.
- Retail participants purchase Class B for passive income.
- Traders access Class C for liquid exposure without compliance overhead.

4.4 Asset Type Configurations

The E-RWA framework adapts to different asset classes through configuration parameters:

Table 4. E-RWA configuration by asset class.

Asset Class	Yield Source	Valuation	Custody
Real Estate	Rental income	Appraised value	Title deed (ERC-1643)
Gold	None (appreciation)	Spot price oracle	Vault certificate
Commodities	Spot/futures spread	Market price	Warehouse receipt
IP / Royalties	Royalty streams	DCF model	License agreement
Private Credit	Interest payments	Face value	Loan agreement

4.5 Formal Token Model

Let $\mathcal{A}$ denote the set of tokenized assets. For each asset $a \in \mathcal{A}$, the E-RWA token system is defined as:

$$T_a = (S_a, P_a, D_a, \Sigma_a) \quad (1)$$

where $S_a = \{s_a^A, s_a^B, s_a^C\}$ is the supply across partitions, P_a is the set of compliance predicates, D_a is the document registry, and Σ_a is the set of authorized signers (controller addresses).

A transfer of n tokens of class k from address α to β is valid if and only if:

$$\text{transfer}(a, k, \alpha, \beta, n) \Leftrightarrow \forall p \in P_a^k : p(\alpha, \beta, n) = 1 \quad (2)$$

where P_a^k is the set of compliance predicates applicable to class k of asset a . This formalization ensures that compliance is composable and asset-specific.

5. Tokenization Flow

The tokenization lifecycle proceeds through four phases, each involving distinct protocol facets and participants (Figure 7).

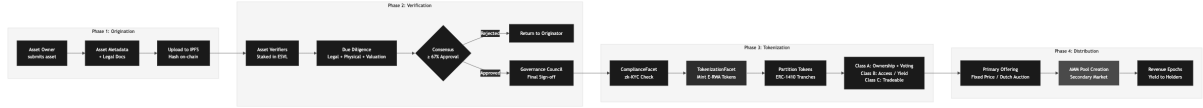


Figure 7. End-to-end tokenization flow. An asset progresses through Origination (metadata and legal docs uploaded to IPFS), Verification (staked verifiers conduct due diligence, governance approves), Tokenization (compliance check, E-RWA minting with Class A/B/C partitions), and Distribution (primary offering followed by AMM secondary market and revenue epochs).

5.1 Phase 1: Asset Origination

An Asset Originator submits an asset for tokenization by providing:

1. **Asset metadata:** Type, description, location (if physical), valuation basis, and yield characteristics.
2. **Legal documentation:** Title deeds, ownership certificates, assay reports, appraisals, or license agreements. Uploaded to IPFS/Arweave with content hashes stored on-chain via the DocumentFacet.
3. **Tokenization parameters:** Requested total supply, class distribution (A/B/C ratios), target jurisdictions, and proposed compliance rules.
4. **Origination bond:** A refundable ESVL deposit (minimum 10,000 ESVL) that is returned upon successful tokenization or forfeited if the asset fails verification due to fraud or misrepresentation.

The DocumentFacet hashes and anchors all documents on-chain:

$$D_a = \{(h_i, \text{uri}_i, t_i, \sigma_i)\}_{i=1}^m \quad (3)$$

where $h_i = \text{keccak256}(\text{doc}_i)$, uri_i is the IPFS/Arweave URI, t_i is the timestamp, and σ_i is the originator's signature.

5.2 Phase 2: Verification

The VerifierFacet assigns the asset to a panel of staked Asset Verifiers. The verification process comprises:

1. **Legal due diligence:** Verifying ownership records, encumbrances, and legal standing.
2. **Physical inspection:** For physical assets (real estate, gold), on-site verification of existence, condition, and custody arrangements.
3. **Valuation:** Independent appraisal aligned with the asset's valuation methodology.

4. **Attestation:** Each verifier submits a signed attestation $A_v = (\text{assetId}, \text{verdict}, \text{findings}, \sigma_v)$ to the VerifierFacet.

Consensus is reached when:

$$\frac{\sum_{i=1}^n w_i \cdot C(v_i, a)}{\sum_{i=1}^n w_i} \geq Q_a \quad (4)$$

where w_i is the stake-weighted influence of verifier v_i , $C(v_i, a) \in \{0, 1\}$ is the verification verdict, and $Q_a \in [0.67, 0.75]$ is the asset-class-specific quorum threshold.

Upon consensus, the Governance Council (veESVL holders) performs a final sign-off vote before the asset proceeds to tokenization.

5.3 Phase 3: Tokenization (Minting)

The ComplianceFacet validates that the originator and proposed token structure comply with applicable jurisdictional rules. The TokenizationFacet then:

1. Creates the E-RWA token contract with configured partitions.
2. Mints tokens according to the approved supply and class distribution.
3. Registers the asset in the global registry within AppStorage.
4. Links all verified documents via the DocumentFacet.

5.4 Phase 4: Distribution and Yield

5.4.1 Primary Distribution

The MarketFacet supports two primary offering mechanisms:

- **Fixed-price offering:** Tokens sold at a predetermined price per unit, suitable for stable-value assets (real estate, bonds).
- **Dutch auction:** Price decreases from a ceiling until demand clears supply, suitable for assets with uncertain market pricing (art, rare commodities).

5.4.2 Secondary Market

After primary distribution, the MarketFacet creates AMM liquidity pools for Class C tokens, enabling continuous trading and price discovery. Class A and B tokens trade on a compliance-gated order book where both parties must satisfy the ComplianceFacet's transfer predicates.

5.4.3 Revenue Distribution

The RevenueFacet collects yield generated by tokenized assets and distributes it to E-RWA holders during defined epochs:

$$r_{a,i} = R_a \cdot \frac{b_{a,i}}{\sum_j b_{a,j}} \quad (5)$$

where R_a is the total revenue collected for asset a in the epoch, and $b_{a,i}$ is the token balance of holder i (across eligible classes A and B).

6. Community Governance

Esvil Protocol governance is fully community-driven, with no admin keys or centralized control after deployment. All protocol decisions flow through the GovernanceFacet (Figure 8).

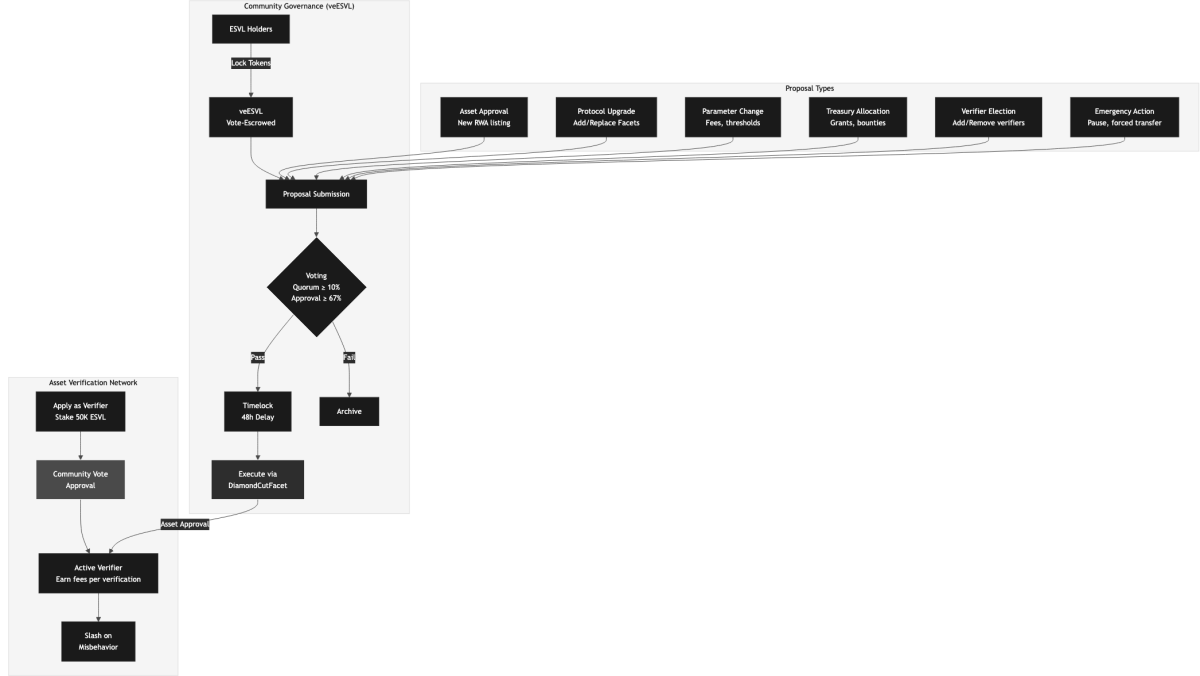


Figure 8. Governance and verification flow. ESVL holders lock tokens to obtain veESVL voting power. Six proposal types cover asset approvals, protocol upgrades, parameter changes, treasury allocation, verifier elections, and emergency actions. The Asset Verification Network operates through staked verifiers elected by community vote.

6.1 Vote-Escrowed ESVL (veESVL)

Governance power is obtained by locking ESVL tokens for a chosen duration, receiving veESVL in return. Longer lock periods yield higher voting multipliers:

Table 5. veESVL lock periods and voting multipliers.

Lock Period	Multiplier	Decay
3 months	1.0×	Linear
6 months	1.5×	Linear
12 months	2.5×	Linear
24 months	5.0×	Linear
48 months	10.0×	Linear

veESVL voting power decays linearly toward the unlock date, incentivizing re-locking and long-term alignment:

$$\text{veESVL}_i(t) = S_i \cdot m_i \cdot \frac{t_{\text{unlock}} - t}{t_{\text{unlock}} - t_{\text{lock}}} \quad (6)$$

where S_i is the locked amount, m_i is the lock multiplier, and t is the current time.

6.2 Proposal Types

Six categories of governance proposals are supported:

1. **Asset Approval:** Approve or reject a verified asset for tokenization.
2. **Protocol Upgrade:** Add, replace, or remove Diamond facets via `diamondCut()`.
3. **Parameter Change:** Adjust fees, quorum thresholds, slashing rates, epoch durations.
4. **Treasury Allocation:** Deploy treasury funds for grants, bounties, or ecosystem development.
5. **Verifier Election:** Add or remove Asset Verifiers from the active set.
6. **Emergency Action:** Pause protocol, execute forced transfers (ERC-1644), or deploy critical patches.

Standard proposals require 10% quorum and 67% approval. Emergency proposals require 5% quorum and 80% supermajority with a shortened 24-hour timelock.

6.3 Decentralized Asset Verification Network

Asset Verifiers are independent entities responsible for due diligence on proposed assets. The network operates as follows:

1. **Application:** Prospective verifiers stake a minimum of 50,000 ESVL and submit credentials.
2. **Election:** veESVL holders vote to approve or reject the applicant.
3. **Operation:** Active verifiers are assigned to asset verification panels, earning fees per completed verification.
4. **Accountability:** Verifiers who submit fraudulent or negligent attestations face slashing penalties as defined in Table 6.

Table 6. Verifier slashing schedule.

Offense	Slash
Negligent attestation (first)	15% of stake
Negligent attestation (repeat)	35% of stake
Fraudulent attestation	100% of stake + removal
Collusion with originator	100% + permanent ban

Slashed tokens are burned permanently, creating deflationary pressure and deterring misbehavior.

7. Token Economics

The ESVL token is the native BEP-20 utility and governance token powering all protocol operations (Figure 9).

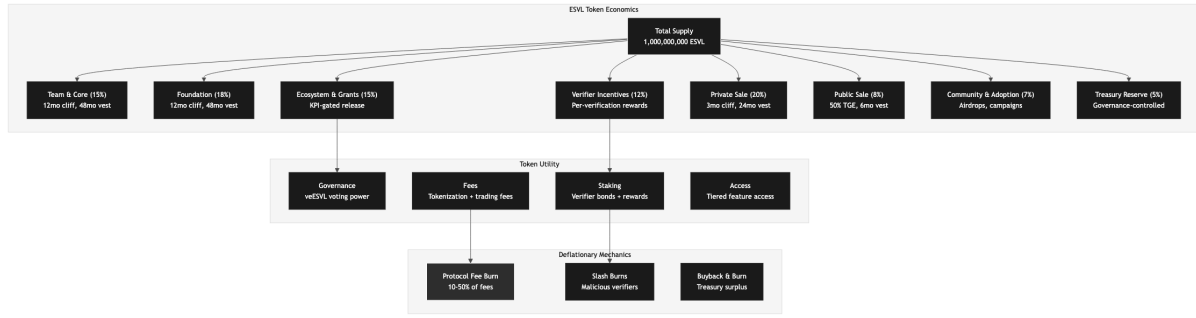


Figure 9. ESVL token economics. Total supply of 1B tokens distributed across eight categories with vesting schedules. Token utility spans governance (veESVL), staking (verifier bonds), fees (tokenization and trading), and feature access. Deflationary mechanics include protocol fee burns, slash burns, and treasury buyback-and-burn programs.

7.1 Token Utility

Table 7. ESVL token utility domains.

Domain	Function
Governance	Lock for veESVL, vote on all protocol decisions
Staking	Verifier bonds, liquidity incentives, protocol security
Fees	Tokenization fees, trading fees, verification fees (partial burn)
Access	Tiered feature access based on staked amount
Origination	Refundable bonds for asset submission

7.2 Token Allocation

Initial supply: 1,000,000,000 ESVL.

Table 8. ESVL token allocation and vesting.

Category	%	Cliff	Vest	TGE
Team & Core	15	12 mo	48 mo	0%
Foundation	18	12 mo	48 mo	0%
Ecosystem & Grants	15	0	48 mo	–
Verifier Incentives	12	12 mo	48 mo	0%
Private Sale	20	3 mo	24 mo	10%
Public Sale	8	0	6 mo	50%
Community & Adoption	7	0	36 mo	5%
Treasury Reserve	5	–	Gov.	0%

7.3 Goal-Oriented Distribution

Ecosystem and Grant allocations follow a goal-oriented model where token emissions correlate with protocol performance. Let θ_t denote the current KPI state and θ_t^* the target:

$$a_t = \rho(t, \theta_t) \cdot \Delta S_t \quad (7)$$

where

$$\rho(t, \theta_t) = \rho_m + (\rho_M - \rho_m) \cdot d(\theta_t, \theta_t^*) \quad (8)$$

The composite distance function weights five KPIs:

Table 9. KPI weights for goal-oriented distribution.

KPI	Weight
Total Value Tokenized (TVT)	0.30
Active Verifiers	0.20
Unique Token Holders	0.20
Trading Volume	0.15
Governance Participation	0.15

7.4 Deflationary Mechanics

Three burning mechanisms counterbalance supply growth:

1. **Protocol fee burn:** A governance-adjustable percentage of all protocol fees is burned (10% in Year 1, scaling to 50% by Year 3).
2. **Slash burns:** All slashed tokens from malicious verifiers are burned permanently.
3. **Treasury buyback-and-burn:** When treasury reserves exceed 150% of the target balance, surplus is used to buy and burn ESVL from the open market.

Supply evolution follows:

$$S_{t+1} = S_t + M_t - B_t^{\text{fee}} - B_t^{\text{slash}} - B_t^{\text{buyback}} - L_t + U_t \quad (9)$$

where M_t is minted tokens, B_t terms are burns, and L_t, U_t are governance-locked and unlocked tokens respectively.

Proposition. After all allocated tokens are minted ($M_t = 0$ for $t \geq \tau$), the expected supply change is negative, creating long-term deflationary pressure.

Proof. For $t \geq \tau$:

$$\mathbb{E}[S_{t+1} - S_t] = -(B_t^{\text{fee}} + B_t^{\text{slash}} + B_t^{\text{buyback}}) - \mathbb{E}[L_t - U_t]$$

All burn terms are positive, and assuming average governance locking exceeds unlocking ($\mathbb{E}[L_t] \geq \mathbb{E}[U_t]$), the expected supply change is strictly negative. $\square$

8. Regulatory Compliance

8.1 Composable Compliance Modules

The ComplianceFacet implements a pluggable compliance engine where jurisdiction-specific rules are encoded as modular compliance agents:

```
interface IComplianceModule {
    function canTransfer(
        address from,
        address to,
        uint256 amount,
        bytes32 partition
    ) external view returns (bool);

    function transferred(
        address from,
        address to,
        uint256 amount,
        bytes32 partition
    ) external;
}
```

Listing 2. Compliance module interface

Multiple compliance modules can be composed per asset. A transfer is valid only if *all* applicable modules approve:

$$\text{valid}(\alpha, \beta, n, k) = \bigwedge_{j=1}^m M_j.\text{canTransfer}(\alpha, \beta, n, k) \quad (10)$$

8.2 Supported Compliance Rules

Table 10. Example compliance module types.

Module	Rule
MaxHolders	Limits total number of token holders per asset
AccreditedOnly	Restricts Class A to accredited/institutional investors
JurisdictionBlock	Blocks transfers to/from specified jurisdictions
HoldingPeriod	Enforces minimum holding periods before transfer
MaxOwnership	Caps maximum percentage ownership per holder
TransferWindow	Restricts transfers to defined time windows

8.3 zk-KYC and ONCHAINID

Identity verification uses privacy-preserving zero-knowledge proofs:

1. Users complete KYC with an authorized verification provider off-chain.

2. The provider issues a zk-proof attesting to identity attributes (jurisdiction, accreditation status) without revealing personal data.
3. The proof is registered in the ONCHAINID Registry, linked to the user's wallet.
4. The ComplianceFacet queries ONCHAINID during transfer validation.

This approach satisfies AML/KYC requirements while preserving user privacy, a critical consideration under GDPR and similar data protection frameworks.

8.4 Jurisdictional Strategy

Table 11. Initial target jurisdictions and regulatory frameworks.

Jurisdiction	Framework	Asset Focus
Nigeria	SEC Nigeria Digital Assets Rules	Real estate, commodities
Switzerland	DLT Act (2021)	Tokenized securities
Estonia	Virtual Asset Service Provider License	Digital asset custody
Singapore	MAS Payment Services Act	Tokenized funds
Dubai (DIFC)	DFSA Crypto Token Regime	Broad RWA coverage
Liberland	Liberland Blockchain Laws; LLM token framework	Tokenized governance, digital assets
Burkina Faso	Code Minier 2015 (revised); SONASP partnership model	Gold, mineral commodities

8.4.1 Liberland: Blockchain-Native Sovereignty

The Free Republic of Liberland presents a unique jurisdictional opportunity as a blockchain-native microstate. Liberland's governance infrastructure is built on distributed ledger technology, with its Liberland Merit (LLM) token serving as both a governance instrument and a unit of account. The state's legal framework explicitly recognizes tokenized assets and smart contract-based governance, making it one of the few jurisdictions where the Esvil Protocol's on-chain governance model (veESVL voting, Diamond upgrades, verifier elections) operates within a natively compatible legal environment. Liberland's openness to decentralized autonomous structures and its minimal regulatory overhead make it an ideal sandbox for pioneering novel RWA tokenization models before scaling to larger jurisdictions.

8.4.2 Burkina Faso: Tokenized Gold and Local Economic Empowerment

Burkina Faso is Africa's fourth-largest gold producer, with artisanal and small-scale mining (ASM) accounting for a significant share of national output. The *Société Nationale de Substances Précieuses* (SONASP), the state agency responsible for purchasing, refining, and marketing artisanal gold, provides a natural institutional partner for tokenized gold provenance and trade. Under the revised *Code Minier* (2015), Burkina Faso has progressively formalized its mining sector, creating regulatory pathways for technology-enabled traceability and value-chain transparency.

The Esvil Protocol's integration with Burkina Faso targets three objectives:

1. **Tokenized marked gold:** Artisanally mined gold, once assayed and marked by SONASP-accredited refineries, is tokenized as E-RWA tokens. The DocumentFacet anchors assay certificates, chain-of-custody records, and SONASP purchase receipts on-chain, creating an immutable provenance trail from mine site to market.
2. **Local economic participation:** Gold miners and cooperatives receive Class B E-RWA tokens representing fractional ownership in refined gold inventories, enabling them to participate in downstream value creation rather than selling raw output at discounted pit-head prices.
3. **Sovereign gold reserves on-chain:** In partnership with SONASP, tokenized gold reserves can serve as collateral for decentralized lending, unlocking liquidity for local infrastructure development while maintaining sovereign custody of physical assets.

This model directly addresses the value leakage that characterizes artisanal gold supply chains in West Africa, where intermediaries capture disproportionate margins. By tokenizing marked gold at the point of SONASP certification, the protocol ensures transparent pricing, reduces counterfeiting risk, and channels value back to producing communities.

9. Use Cases

9.1 Real Estate: Fractional Commercial Ownership

An office building valued at \$2M is tokenized into 2,000,000 E-RWA tokens. Class A holders (accredited investors) vote on tenant selection and capital expenditures. Class B holders receive monthly rental yield. Class C tokens trade freely on the AMM, providing liquid exposure.

9.2 Gold: Tokenized Vault Certificates

A 100kg gold deposit in an audited vault is tokenized with each E-RWA token representing 0.001kg. DocumentFacet stores the vault certificate and quarterly audit reports. Verifiers include the vault operator and an independent auditor. No yield distribution; value tracks spot price via oracle integration.

9.3 MSME Physical Spaces

The protocol's original use case: MSMEs pool capital to co-own commercial spaces. 50 MSMEs each acquire Class B tokens representing workspace access rights and pro-rata revenue from subleasing unused capacity. Governance (Class A) is held by the community cooperative.

9.4 Intellectual Property and Royalties

A music catalog generating \$50K/month in streaming royalties is tokenized. Class B holders receive monthly yield distributions through RevenueFacet epochs. The copyright holder retains Class A governance rights over licensing decisions.

9.5 Agricultural Commodities

A 500-ton cocoa harvest in bonded warehouse is tokenized against warehouse receipts. DocumentFacet stores the receipt, quality certificates, and insurance documentation. Tokens are sold via Dutch auction, with delivery settlement coordinated off-chain through an authorized custodian.

9.6 Burkina Faso: SONASP-Certified Artisanal Gold

A cooperative of 200 artisanal miners in the Sahel region produces 50kg of gold per quarter. The gold is assayed, marked, and certified by a SONASP-accredited refinery. Each kilogram is tokenized into 1,000 E-RWA tokens with full provenance documentation (mine coordinates, assay report, SONASP receipt, chain-of-custody log) anchored via the DocumentFacet. Class A governance is held by the cooperative's elected council. Class B tokens distribute quarterly revenue from gold sales to member miners proportionally. Class C tokens enable international investors to gain liquid exposure to Burkinabè artisanal gold at transparent, oracle-verified spot pricing, bypassing opaque intermediary networks.

10. Security and Risk Assessment

10.1 Smart Contract Security

- **Multiple audits:** Minimum three tier-1 audit firms before mainnet launch.
- **Formal verification:** Critical facets (Tokenization, Compliance, Governance) undergo formal verification.
- **Bug bounty:** Ongoing program with rewards up to \$500K for critical vulnerabilities.
- **Emergency pause:** GovernanceFacet includes circuit breaker functionality, activatable by supermajority vote.
- **Insurance fund:** 5% of protocol revenue allocated to a smart contract cover fund.

10.2 Diamond-Specific Risks

Table 12. Diamond pattern risk assessment.

Risk	Description	Prob.	Mitigation
Selector clash	Two facets register the same function selector	Low	Automated clash detection in DiamondCutFacet
Storage collision	Facets overwrite shared storage slots	Low	Single AppStorage struct, no per-facet storage
Malicious upgrade	Governance compromised, bad facet deployed	Low	Timelock + multisig + supermajority requirement
Facet reentrancy	Cross-facet delegatecall reentrancy	Med	ReentrancyGuard in AppStorage, CEI pattern

10.3 Economic Risk Matrix

Table 13. Protocol economic risk assessment.

Category	Risk	Prob.	Impact
Verification	Verifier collusion or capture	Low	Critical
Market	Insufficient secondary market liquidity	Med	High
Economic	ESVL token price collapse	Med	Medium
Legal	Asset ownership disputed off-chain	Low	High
Adoption	Low originator demand	Med	High
Regulatory	Broad securities classification	Med	High

11. Implementation Roadmap

Table 14. Phased implementation roadmap.

Phase	Milestones
Phase I (Q2 2026)	Diamond proxy and core facets deployed to BNB Testnet; ESVL token contract; zk-KYC integration; governance voting
Phase II (Q3 2026)	First asset tokenized (real estate pilot, Nigeria); verifier network launched; DocumentFacet with IPFS
Phase III (Q4 2026)	AMM pools for Class C tokens; revenue distribution epochs; second asset class (gold)
Phase IV (Q1 2027)	Mainnet launch on BNB Chain; multi-jurisdiction compliance modules; third asset class
Phase V (2027+)	Cross-chain bridges; institutional partnerships; additional jurisdictions; advanced financial products (collateralized positions, derivatives)

12. Conclusion

The tokenization of real-world assets represents one of the most significant opportunities in decentralized finance, yet the infrastructure to support it at scale remains fragmented, rigid, and insufficiently compliant. The Esvil Protocol addresses these limitations through three architectural innovations: the ERC-2535 Diamond Pattern enables modular, upgradeable infrastructure behind a single contract address; the E-RWA hybrid token standard provides a composable framework adaptable to any asset class; and community governance through veESVL ensures that protocol evolution reflects the collective interests of its participants.

By making RWA tokenization asset-agnostic, compliance-aware, and community-governed, Esvil creates the conditions for a genuinely open market where real estate, precious metals, commodities, intellectual property, and any other value-bearing asset can be fractionalized, traded, and composed within a unified decentralized ecosystem on BNB Chain.

The protocol invites asset originators, verifiers, investors, developers, and governance participants to build the infrastructure that bridges traditional value and decentralized finance, one asset at a time.

References

- Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System.
- Buterin, V. (2014). Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform. *Ethereum Whitepaper*.
- Wood, G. (2014). Ethereum: A Secure Decentralised Generalised Transaction Ledger. *Ethereum Yellow Paper*.

- Mudge, N. (2020). ERC-2535: Diamonds, Multi-Facet Proxy. *Ethereum Improvement Proposals*.
- T-REX Protocol (2021). ERC-3643: T-REX Token for Real Estate Exchange. *Ethereum Improvement Proposals*.
- Edmonds, F., et al. (2018). ERC-1400 Security Token Standard (ERC-1410, ERC-1643, ERC-1644). *Ethereum Improvement Proposals*.
- Schär, F. (2021). Decentralized Finance: On Blockchain- and Smart Contract-Based Financial Markets. *Federal Reserve Bank of St. Louis Review*, 103(2):153–174.
- Zetzsche, D. A., Buckley, R. P., & Arner, D. W. (2020). Decentralized Finance (DeFi). *Journal of Financial Regulation*, 6(2):172–201.
- McKinsey & Company (2024). Tokenization: A Digital Asset Déjà Vu. *McKinsey Global Institute Report*.
- World Bank (2023). Global Economic Prospects: Africa’s Pulse. *World Bank Publications*.
- International Monetary Fund (2023). World Economic Outlook: Navigating Global Divergences. *IMF Publications*.
- OECD (2022). Tokenisation of Assets and Financial Instruments. *OECD Publishing*.
- OECD (2022). SME and Entrepreneurship Outlook 2022. *OECD Publishing*.
- OpenZeppelin (2024). Contracts Documentation: Governor, ERC-20, Diamond Patterns. <https://docs.openzeppelin.com/>.
- Zyskind, G., Nathan, O., & Pentland, A. (2015). Decentralizing Privacy: Using Blockchain to Protect Personal Data. *2015 IEEE Security and Privacy Workshops*, 180–184.
- Lamport, L., Shostak, R., & Pease, M. (1982). The Byzantine Generals Problem. *ACM TOPLAS*, 4(3):382–401.
- BNB Chain Documentation (2024). BNB Smart Chain: Technical Overview and Developer Resources. <https://docs.bnbchain.org/>.
- Szabo, N. (1997). Formalizing and Securing Relationships on Public Networks. *First Monday*, 2(9).
- Joshi, S. & Choudhury, A. (2022). Tokenization of Real Estate Assets Using Blockchain. *International Journal of Intelligent Information Technologies*, 18(3):373–393.
- Konashevych, O. (2020). General Concept of Real Estate Tokenization on Blockchain: The Right to Choose. *Preprint, Erasmus Mundus LAST-JD*. DOI: 10.13140/RG.2.2.33435.62244.
- Juan, A. A., Perez-Bernabeu, E., Li, Y., Martin, X. A., Ammouriova, M., & Barrios, B. B. (2023). Tokenized Markets Using Blockchain Technology: Exploring Recent Developments and Opportunities. *Information*, 14(6):347.

- Norta, A., Udokwu, C., & Craß, S. (2024). Real-World Asset Identity Authentication in Blockchain Enabled Inter-Organizational Process-Aware Systems Involving Adjustable Challenge-Response Evaluation Sets. *SN Computer Science*, 5:936.
- Karamachoski, J., Marina, N., & Taskov, P. (2020). Blockchain-Based Application for Certification Management. *Tehnicki glasnik*, 14(4):488–492.
- Amey Navarro, L. (2024). Decentralized Finance as a Tool for Financial Empowerment: A MSMEs Colombia Case Study. *Bachelor Thesis, Haute école de gestion de Genève (HEG-GE)*.
- Riggs, W., Vyas, V., & Sethi, M. (2022). Blockchain and Distributed Autonomous Community Ecosystems: Opportunities to Democratize Finance and Delivery of Transport, Housing, Urban Greening and Community Infrastructure. *Mineta Transportation Institute Publications*, Project 2165.
- Akaba, T. I., Norta, A., Udokwu, C., & Draheim, D. (2020). A Framework for the Adoption of Blockchain-Based e-Procurement Systems in the Public Sector: A Case Study of Nigeria. In *Responsible Design, Implementation and Use of ICT*, LNCS 12066, pp. 3–14. Springer.
- Udokwu, C. (2024). Formalizing and Simulating the Token Aspects of Blockchain-Based Research Collaboration Platform Using Game Theory. *Mathematics*, 12(20):3252.
- Akbar, D., Ünal, M., Akkaya, R., & Balci, O. (2025). Blockchain-Based Peer-to-Peer Energy Trading: Design and Implementation of the Tokenized Decentralized Energy Management System (DEMS). *Preprint*. DOI: 10.21203/rs.3.rs-6787257/v1.
- Baldimtsi, F. & Cachin, C., Eds. (2023). *Financial Cryptography and Data Security: 27th International Conference, FC 2023*. LNCS 13950, Springer.
- Benton, M. C. & Radziwill, N. M. (2017). Quality and Innovation with Blockchain Technology. *arXiv preprint*, 1710.04130.